

Coding & Solution

Date	4 October 2026
Team ID	6ab2951f1266c775575a6b80
Project Name	AI-Powered Smart Agri-Monitor
Maximum Marks	10 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/rajagopalrg946-maker/Farm-Management-System.git
Programming Language(s)	Python, JavaScript (React), SQL
Framework(s) Used	Flask (Backend), React (Frontend), TensorFlow, PostgreSQL
Key Features Implemented	Real-time soil sensor data ingestion, Weather pattern prediction using AI, Automated irrigation alert system, Crop health dashboard with heatmaps
Pending / Incomplete Features	Mobile app integration, Historical yield trend analysis, Multi-language support
Setup / Run Instructions	1. Install dependencies (pip install -r requirements.txt, npm install). 2. Set up DB (PostgreSQL). 3. Start Flask server. 4. Run React frontend. 5. Detailed guide in repository README.

Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes (e.g., more detailed docstrings for AI modules needed)	Partial
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The core AI models are implemented. Data from sensors is processed. Code base is well-structured and follows modularity. Some additional time is needed to polish the user interface and integrate historical data analysis. A preliminary user guide is included.